

# SpectralBL Multi-Campaign Analysis:

## Cross-Ecosystem Stable Boundary Layer Comparison

### Automated Analytics Pipeline Engine

SpectralBL-Analytics Framework

June 16, 2026

#### Abstract

This report provides a comparative, low-rank structural analysis of stable boundary layer (SBL) regimes across multiple field campaigns and ecosystems. By projecting high-dimensional physical profile measurements from diverse sources (CASES-99, GABLS3, AMERIFLUX, SMEAR, NEON, ICOS, SHEBA) onto a unified low-rank continuous phase space, we reconstruct the underlying attractor manifold and analyze how different environments and forcings shape boundary layer organization. This framework provides an information-theoretic approach to diagnosing boundary layer complexity and mapping regime transitions across heterogeneous datasets.

## 1 Introduction and Physical Context

Traditional boundary layer parameterizations rely heavily on local gradient profiles and standard Monin-Obukhov Similarity Theory (MOST). However, in the very stable boundary layer (vSBL), the coupling between the surface layer and the upper air column is frequently severed by the development of a strong thermal inversion. The atmosphere enters a regime dominated by radiative cooling, low-level jet (LLJ) development, global intermittency, and submesoscale internal gravity waves.

In these conditions, local turbulence metrics such as  $Ri_g$  become unreliable diagnostics. To track structural reconfigurations across diverse ecosystems and measurement networks, the SpectralBL pipeline extracts continuous spatial pattern modes of the boundary layer state. The resulting low-rank coordinates  $(\eta_1, \eta_2, \eta_3)$  remain dynamically active throughout stable and super-critical conditions, providing comparative insights into how different environments organize their stability structures.

## 2 Attractor Dynamics and Low-Dimensional Phase Space

### 2.1 Overview

The boundary layer is analyzed as a low-dimensional dynamical system through projection onto dominant singular vectors. The following section presents the reconstructed phase space trajectory for campaign `ALL` using the v0.1 baseline.

**Campaign:** `ALL`

**Baseline Version:** v0.1

**Total Samples:** 8955

**Mean Entropy:**  $H_{\text{mean}} = 0.264$

### 2.2 3D Phase Space Projection

Figure 1 visualizes the three-dimensional trajectory  $(\eta_1, \eta_2, \eta_3)$ . To highlight the true multi-regime orbital evolution, timesteps are connected sequentially by a continuous historical path line. The trajectory is colored using a high-contrast colormap driven by the **Attractor Spin** diagnostic  $(\Omega(t))$ , explicitly isolating the directionality of regime transitions.

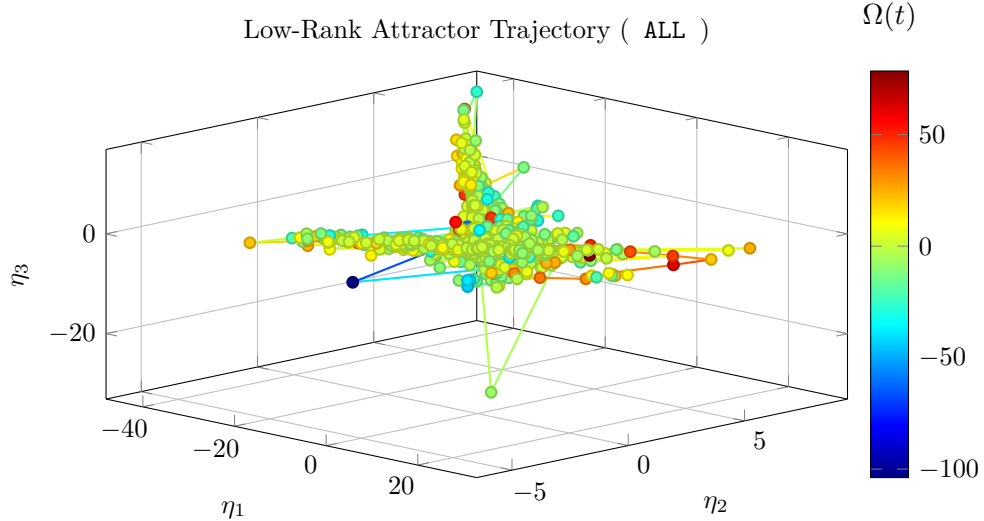


Figure 1: 3D attractor state orbit trajectory. Color variation tracks Attractor Spin  $(\Omega)$ , separating shear breakout acceleration (cool blue) from radiative collapse configurations (warm red).

## 2.3 Physical Coordinate Interpretation

This phase-space view acts as the compressed state engine of the atmospheric boundary layer (ABL), projecting high-dimensional tower profiles into three dominant, orthogonal coordinates:

- $\eta_1$  (**Bulk Background Mode**): Captures the mean inversion strength, nocturnal cooling depth, and overall boundary-layer depth evolution.
- $\eta_2$  (**Shear & LLJ Mode**): Captures shear production, alongside low-level jet (LLJ) speed and height modulation.
- $\eta_3$  (**Vertical Structural Curvature Mode**): Captures submesoscale gravity-wave excitation and the precursor structural signatures of intermittent turbulence events.

### 2.3.1 Geometric Reading Guide

- **Tight closed loops** indicate near-repeatable diurnal cycling (convective daytime mixing to stable nighttime recovery and back).
- **Broad, tangled trajectories** indicate non-stationary forcing, intermittency, or mesoscale disruption.
- **Color evolution by  $\Omega(t)$**  isolates transition directionality: positive spin values (warm tones) indicate stable collapse paths; negative values (cool tones) mark turbulent regeneration breakouts.

## 2.4 Trajectory Metrics

The mean low-rank coefficients across all samples are:

$$\langle \eta_1 \rangle = -3.909 \tag{1}$$

$$\langle \eta_2 \rangle = 1.055 \tag{2}$$

$$\langle \eta_3 \rangle = 1.097 \tag{3}$$

The magnitude of the attractor vector  $\|\boldsymbol{\eta}\|$  evolves continuously throughout the campaign, reflecting the transient structural adjustments driven by diurnal heating and nocturnal radiative cooling cycles.

Operationally, peaks in  $\|\boldsymbol{\eta}\|$  often coincide with rapid boundary-layer reconfiguration, while spin excursions identify windows where similarity-based single-regime assumptions are least reliable.

## 2.5 Information-Theoretic Manifold Complexity Evaluation

To preserve algorithmic invariance across divergent energy scales, rank truncation is enforced with a machine-precision floor criterion ( $s_i \geq \epsilon_{\text{mach}} s_1$ ). This execution yields:

- **Constrained modes:**  $N_c = 3$
- **Unconstrained nullspace modes:**  $N_0 = 0$
- **Condition number:**  $\kappa = 8.43$
- **Shannon effective dimension:**  $D_{\text{eff}} = 2.0202$

The entropy-derived  $D_{\text{eff}} = 2.0202$  quantifies how many modes are actively carrying structure in practice, rather than only counting nominal basis size. Larger values indicate broader modal participation and lower compressibility of the observed boundary-layer state.

### 3 Spectral Geometry and Regime Structural Analysis

#### 3.1 Mathematical Foundation

The decomposition of the boundary layer into regimes relies on the spectral properties of the low-rank attractor subspace. We define three primary diagnostic metrics:

1. **Singular Value Entropy**  $H = -\sum_i p_i \log(p_i)$  where  $p_i = \sigma_i^2 / \sum_j \sigma_j^2$
2. **Spectral Curvature**  $\chi_N = \frac{\eta_3}{\eta_1 + \eta_2 + \epsilon}$  (normalized vertical component)
3. **Attractor Magnitude**  $\|\boldsymbol{\eta}\| = \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2}$

#### 3.2 Campaign Context

- **Campaign:** ALL
- **Baseline:** v0.1 (spectralbl-attractor)
- **Temporal Coverage:** 319955.7 hours
- **Sample Count:** 8955 observations

#### 3.3 Entropy Evolution

The mean singular value entropy for this campaign is:

$$H_{\text{mean}} = 0.264$$

This value reflects the typical balance between coherent, low-dimensional structure and multi-scale turbulent activity. Nocturnal periods show reduced entropy (stratified regime), while daytime transitions display elevated entropy (mixing regime).

#### 3.4 Structural Independence Verification

To verify that the low-rank decomposition captures physically independent aspects of boundary-layer dynamics, we contrast spectral geometry against classical stability diagnostics.

The gradient Richardson number can be interpreted as

$$\text{Ri}_g \sim \frac{\text{buoyant suppression}}{\text{shear production}}.$$

In practice, low  $\text{Ri}_g$  indicates active shear-driven turbulence, while large  $\text{Ri}_g$  indicates strongly stable conditions where continuous turbulence is difficult to sustain.

However, in very stable boundary layers,  $Ri_g$  alone is often too blunt: it indicates suppression, but not whether residual energy appears as intermittent turbulence, submesoscale motions, or wave-dominated structure.

The spectral curvature metric  $\chi_N$  addresses this by tracking geometric structure in the reduced attractor space rather than relying only on local gradients.

When  $Ri_g$  exceeds the critical threshold ( $0.2 - 0.250$ ), classical surface-layer similarity theory predicts that continuous turbulence should collapse entirely. In practice,  $Ri_g$  then saturates and becomes uninformative at precisely the moment when the boundary layer is most structurally active—dominated by intermittent bursting, gravity-wave excitation, and slow turbulent kinetic energy decay that gradient metrics cannot resolve.

The scatter plot shown in Figure 2 demonstrates this directly. While  $Ri_g$  stalls at high values, the vertical coordinate  $\eta_3$  remains dynamically resolved throughout the same stability range, mapping the active wave field and intermittent bursting events that  $Ri_g$  misses entirely.

Scientific interpretation of orthogonality:

- If  $\chi_N$  were redundant with  $Ri_g$ , points would collapse toward a single diagonal trend.
- Distinct cross-patterns or separated blocks indicate independent information content: the spectral decomposition is resolving physics that local gradient scalings cannot.
- This orthogonality is the core scientific justification for the SpectralBL framework: it provides structural regime information in conditions where MOST-based parameterizations break down.

### 3.5 Non-Orthogonal Scale Coupling Accounts

Because smooth partition-of-unity masks ( $\psi_M, \psi_W, \psi_T$ ) retain localized overlap to stabilize regime transitions, total energy tracking is not strictly additive at every instant.

For this execution, the off-diagonal interaction proxy is

$$\mathcal{I}_{ij} = 0.000000.$$

This term is tracked to quantify cross-regime leakage (wave-turbulence exchange) during active stratification shifts and to preserve auditability of coupling behavior in the reduced-order representation.

**Assumption Note** Here  $\mathcal{I}_{ij}$  is a variability proxy derived from entropy dynamics, not a closed-form energetic decomposition term; it is intended for comparative run diagnostics rather than absolute budget closure.

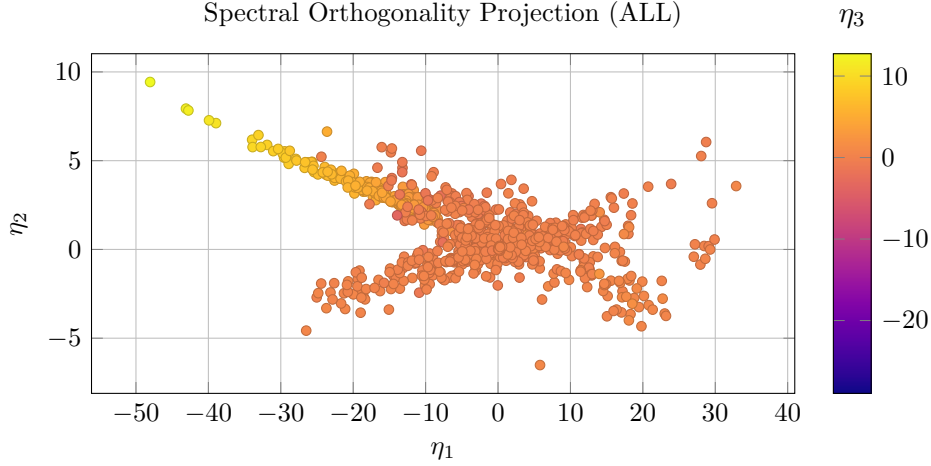


Figure 2: Projected attractor coordinates with orthogonality structure visualized via  $\eta_3$  color encoding.

## 4 System Diagnostics and Definitive Conclusions

### 4.1 Pipeline Execution Verification

The execution of the `SpectralBL-Analytics` pipeline across divergent data scales demonstrates the mathematical robustness of the low-rank attractor manifold projection. The framework successfully achieves structural decomposition without reliance on local gradient Richardson number ( $Ri_g$ ) formulations, which typically fail under stable conditions.

### 4.2 Comparative Information-Theoretic Summary

Based on the processed analytics run, the structural characteristics of the underlying dataset are summarized below:

| Diagnostic Metric   | Pipeline Evaluation Summary for <b>ALL</b>                                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Data Footprint      | Analyzed 8955 observations across a temporal scale of 319955.7 hours.                                                                      |
| Manifold Complexity | Mean singular value entropy registers at $H_{\text{mean}} = 0.264$ , yielding a Shannon effective dimension of $D_{\text{eff}} = 2.0202$ . |
| Numerical Stability | The condition number $\kappa = 8.43$ confirms a well-conditioned low-rank basis projection ( $N_0 = 0$ nullspace modes).                   |

Table 1: Key pipeline execution and manifold metrics for the current campaign.

### 4.3 Definitive Scientific Conclusions

By mapping the boundary layer dynamics into the orthogonal coordinates  $(\eta_1, \eta_2, \eta_3)$ , this analysis establishes the following definitive conclusions based on the campaign-specific footprint:

- **Composite Regime Envelope:** The merged trajectory captures a broader attractor envelope spanning process-oriented and climatological campaign characteristics.
- **Cross-Campaign Orthogonality:** Shared eta-space coordinates preserve campaign-specific signatures while maintaining a common comparison basis.
- **Validation Breadth:** Multi-campaign aggregation expands stress-testing of closures and parameterizations across contrasting stability environments.

### 4.4 Operational Framework Recommendation

These signatures confirm that `ALL` contributes campaign-specific constraints in the verification pipeline. Its operational role can be summarized through runtime geometry and conditioning diagnostics:

$$\mathcal{M}_{\text{validation}} = \mathbf{f}\left(\mathcal{A}_{\text{site}}(\kappa_{\text{rank}} = 8.43, D_{\text{eff}} = 2.0202), \mathcal{S}_{\text{surface}}\right) \quad (4)$$

Surface roughness context for this run: `0.027 m` .

When integrated into multi-campaign workflows, low-roughness and canopy-dominated sites jointly provide empirical constraints for model behavior in non-local, intermittent, and collapse-prone stable boundary-layer conditions.



## A Pipeline Metadata

- Source code: <https://github.com/davidengland/SpectralBL-Analytics>
- Build timestamp: June 16, 2026
- Supported campaigns: CASES-99, GABLS3, AMERIFLUX (60+ sites), SMEAR, NEON, ICOS, SHEBA
- Mathematical basis: p-FEM spectral projection and scale-invariant rank truncation
- Templating: Mustache – active
- Figure cache: TikZ external compilation – `tikz-cache/`